FRANCIS XAVIER ENGINEERING COLLEGE (AUTONOMOUS)

Department of Artificial Intelligence and Data Science

Intelligent Employee Record & Workforce Management System (ERMS)

With Interactive 3D Spatial Visualizations & Role-Based Access Architecture

ZEROTH PROJECT REVIEW (ACADEMIC YEAR 2026-2027)

Date of Review: 14.10.2026 (Wednesday) | Class: III AI&DS;

Team Members: [Student Names & Register Numbers]

Supervisor / Guide: [Faculty Guide Name, Department of AI&DS;]

COMPONENT 2 (10 MARKS)

Problem Statement & Objectives

CHALLENGES IN EXISTING WORKFORCE RECORDKEEPING:

- **Fragmented Records:** Employee dossiers are scattered across manual paper archives and isolated spreadsheets.
- **High Error Proneness:** Manual entry produces conflicting education entries, broken experience records, and duplicate files.
- **Security Vulnerabilities:** Lack of granular role segregation permits unauthorized viewing or accidental modification of sensitive payroll & credentials.
- **Cognitive Fatigue:** Traditional enterprise portals present dense, text-heavy tables with zero responsive visual feedback.

DEFINED PROJECT OBJECTIVES:

✔ **Centralized Digital Archive:** Build an enterprise-grade relational database for profiles, degrees, certifications, and career tenures.
✔ **Role-Based Access Control (RBAC):** Establish distinct security barriers between Employee self-service and Administrative governance.
✔ **Modern 3D Spatial UI:** Embed hardware-accelerated WebGL ('Spectral Drape' & 'Hexagonal Vault') for intuitive, fatigue-free data navigation.
✔ **Audit-Ready Transparency:** Ensure instant lookup and verification for institutional accreditation (NBA/NAAC) and compliance.

COMPONENT 3 (5 MARKS)

Project Scope & UN Sustainable Development Goals (SDGs)

Project Scope & Deployment Horizon:

- Designed for educational institutions, engineering corporations, and research labs requiring verified personnel audits.
- Covers full employee lifecycles: onboarding registration, multi-degree verification, tenure records, and administrative oversight.
- Structured relational schemas designed to feed directly into future AI modules (automated resume matching & turnover forecasting).

SDG 8: Decent Work & Economic Growth	SDG 9: Industry, Innovation & Infrastructure	SDG 4: Quality Education & Credential Verification
Fosters operational workplace transparency, eliminates clerical payroll/tenure errors, and boosts enterprise productivity through automated workflows.	Modernizes outdated institutional record infrastructure using lightweight, cross-platform WebGL & PHP containerized pipelines.	Provides structured, tamper-resistant validation repositories for diplomas, degrees, certifications, and academic qualifications.

COMPONENT 4 (10 MARKS)

Preliminary Literature Survey — Minimum 5 Research Papers

No.	Paper Title & Authors	Journal / Year	Methodology	Key Findings & Gap
1	**Design and Implementation of Employee Information Management System** (Zhang & Liu) **[BASE PAPER]**	J. Phys. Conf. Ser., 2021	B/S Architecture, Apache, MySQL relational backend	Proved scalable CRUD record architectures; lacked modern interactive visual interfaces.
2	**Cloud-Based Human Resource Information Systems: Security Analysis** (Rahman et al.)	IEEE Access, 2022	Multi-tenant cloud architecture with role isolation	Proved security benefits of RBAC; noted latency drawbacks in complex visual portals.
3	**Interactive 3D Web Visualizations for Enterprise Dashboards** (Smith et al.)	ACM CHI, 2023	WebGL / Three.js hardware-accelerated shaders	Spatial 3D backgrounds measurably lower cognitive workload during bulk data processing.
4	**Role-Based Access Control and Data Security in Enterprise Systems** (Kumar & Verma)	Springer LNCS, 2023	RBAC privilege matrix and credential hashing	Demonstrated vulnerabilities of shared login routes; confirmed need for separate gateways.
5	**AI-Driven Predictive Analytics in Human Resource Management** (Al-Radaideh et al.)	Elsevier Procedia, 2024	Supervised ML classification on employee career logs	Emphasized that clean, normalized historical databases are the essential bedrock for AI modeling.

COMPONENT 5

Existing Methods and Their Limitations

EXISTING TRADITIONAL METHODS:

• **Manual Paper Dossiers:** Hard-copy files stored in physical archives. Inefficient retrieval, susceptible to environmental wear, and incapable of rapid search.

• **Unprotected Spreadsheet Ledgers:** Microsoft Excel or Google Sheets. Subject to accidental row overwrites, broken formulas, and zero role-based viewing permissions.

• **Rigid Legacy Enterprise ERPs:** Heavy on-premise proprietary applications with steep licensing costs, complex installation steps, and non-intuitive legacy user experiences.

CRITICAL SYSTEM LIMITATIONS & GAPS:

■ **High Record Inconsistency:** Absence of relational foreign-key integrity permits disconnected education and experience entries.

■ **Security & Access Breaches:** Single-tier login setups risk unauthorized profile tampering or disclosure of private administrative metrics.

■ **No Spatial / Modern UX:** Visual fatigue and monotonous forms increase employee churn and data entry mistakes.

■ **Incompatible with Modern AI:** Disorganized data cannot be readily passed to machine learning pipelines for predictive workforce analytics.

COMPONENT 6 (10 MARKS)

Proposed Solution & Architecture Workflow

LAYER 1: PRESENTATION LAYER (Client Tier)
Responsive Bootstrap SB-Admin-2 Dashboard · WebGL Three.js 'Spectral Drape' 3D Waves · Hexagonal Vault Grid · Smooth Spring Cursor Physics

LAYER 2: AUTHENTICATION & GATEWAY LAYER (Security Tier)
Role-Based Access Control (RBAC) · Stateful PHP Session Verification · Separate Entry Gateways for Staff Self-Service and Administrative Oversight

LAYER 3: APPLICATION & BUSINESS LOGIC LAYER (Controller Tier)
PHP 8.x Engine · Profile CRUD Pipelines · Multi-Degree Academic History Tracker · Chronological Corporate Experience Logs

LAYER 4: PERSISTENCE & DATA STORAGE LAYER (Database Tier)
Relational MySQL / MariaDB Engine · Normalized Tables: employeedetail, empeducation, empexpireince, tbladmin · ACID Transactions

COMPONENT 7 (10 MARKS)

Technologies and Resources Required

FRONTEND TECHNOLOGIES	BACKEND & DATABASE	ENVIRONMENT & TOOLS
• **HTML5 / CSS3 / ES6+:** Modern responsive markup and styles • **Three.js / WebGL:** GPU hardware-accelerated 3D wave draping • **SB-Admin-2 Framework:** Enterprise portal layout • **FontAwesome & Google Fonts:** Clean iconography • **Glassmorphism UI:** High-contrast translucent cards	• **PHP 8.5+ Runtime:** High-performance server-side scripting • **MySQL / MariaDB:** ACID-compliant relational DB engine • **MySQLi Driver:** Secure parameterized database querying • **Session Guards:** Tamper-proof role authentication • **Auto Seeder:** Automated relational schema migration	• **Operating System:** macOS Darwin / Unix Environment • **Bootloader:** Bash one-click startup pipeline (start.sh) • **Package Manager:** Homebrew architecture • **Local Host:** Dedicated port binding on localhost:8000 • **Compatibility:** Tested across Chromium, Safari, and Firefox

COMPONENT 8 (5 MARKS)

Expected Outcomes and Tentative Timeline

EXPECTED DELIVERABLES & OUTCOMES:	TENTATIVE PROJECT TIMELINE:
✔ **Fully Operational Web System:** Deployed and verified locally with zero external cloud dependencies. ✔ **Distinct Portal Separation:** High security isolation between staff profile inputs and admin auditing. ✔ **Reduced Administrative Friction:** Over 70% decrease in manual verification time for employee credentials. ✔ **Engaging 3D Visual Physics:** Fluid mouse parallax and glowing particle meshes preventing UI fatigue.	• **Phase 1 (Aug - Oct 2026):** Problem formulation, literature survey, database schema design, and baseline system deployment. **[COMPLETED]** • **Phase 2 (Oct - Nov 2026):** Zeroth review defense, refinement of multi-degree education and employer experience pipelines. **[CURRENT]** • **Phase 3 (Dec 2026 - Jan 2027):** AI-assisted resume parsing & automated profile completion module integration. • **Phase 4 (Feb - Mar 2027):** Security audit, load testing, comprehensive documentation, and final viva review.

THANK YOU!

Employee Record Management System (ERMS)

Ready for Committee Evaluation, Feedback & Live Demonstration

Running live on: **http://localhost:8000/**

Features live 3D Spectral Drape WebGL Physics & Role-Based Gateways